

2024 Robotics Project 1

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Question Explanation

1. Based on the provided DH model, create a six-axis robotic arm model.

Joint	d(in)	a(in)	α	θ
1	0	120mm	-90°	0°
2	0	250mm	0°	0°
3	0	260mm	0°	0°
4	0	0	-90°	0°
5	0	0	90°	0°
6	0	0	0°	0°

Question Explanation

2. input: joint variables, output: Cartesian point (n, o, a, p) and (x, y, z, ϕ , θ , ψ).

- Test case

```
1 % input_template
2 input_T1 = [0.349,0.349,-0.349,0.349,0.349,0.349];
3 input_T2 = [0.873,0.873,-0.873,0.873,0.873,0.873];
```

- Expected result

```
[n, o, a, p]:
    0.7797   -0.6258   -0.0194   577.8388
   -0.0582   -0.1033    0.9929   210.3161
   -0.6234   -0.7731   -0.1170  -85.5050
         0         0         0         1.0000

output:
    577.8388   210.3161  -85.5050  -51.1183   96.7177   91.1183

[n, o, a, p]:
    0.1707   -0.9695   -0.1759   347.5533
   -0.5626   -0.2425    0.7904   414.1979
   -0.8089   -0.0360   -0.5868  -191.5111
         0         0         0         1.0000

output:
    347.5533   414.1979  -191.5111   -2.5463  125.9320  102.5463
```

Question Explanation

3. input: Cartesian point (n, o, a, p), output: the corresponding joint variables. Need to calculate eight sets of solutions and indicate which ones are out of range.

- Test case

$$A1 = \begin{bmatrix} 0.5756 & -0.2398 & -0.7817 & 177.8 \\ 0.7738 & -0.1494 & 0.6156 & 308 \\ -0.2644 & -0.9593 & 0.0996 & -140.1 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix};$$

$$A2 = \begin{bmatrix} 0.1736 & -0.0000 & -0.9848 & 0.0000 \\ 0.8529 & 0.5000 & 0.1504 & 325.2 \\ 0.4924 & -0.8660 & 0.0868 & -158 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix};$$

- Expected Result (Solutions that exceed the working space should be marked as invalid.)

關節角度 (rad) :

60.000 -28.526 115.000 -121.474 10.000 50.000
joint 3 4 out of range

60.000 -28.526 115.000 58.526 -10.000 -130.000
joint 3 out of range

60.000 90.000 -115.000 -10.000 10.000 50.000

60.000 90.000 -115.000 170.000 -10.000 -130.000
joint 4 out of range

-120.000 149.784 27.062 38.154 170.000 -130.000
joint 2 3 out of range

-120.000 149.784 27.062 -141.846 -170.000 50.000
joint 2 3 4 out of range

-120.000 177.387 -27.062 64.675 170.000 -130.000
joint 2 out of range

-120.000 177.387 -27.062 -115.325 -170.000 50.000
joint 2 4 out of range

關節角度 (rad) :

90.000 -23.813 119.000 -125.187 10.000 0.000
joint 3 4 out of range

90.000 -23.813 119.000 54.813 -10.000 -180.000
joint 3 out of range

90.000 99.000 -119.000 -10.000 10.000 0.000

90.000 99.000 -119.000 170.000 -10.000 -180.000
joint 4 out of range

-90.000 137.866 44.277 27.857 170.000 -180.000
joint 2 3 out of range

-90.000 137.866 44.277 -152.143 -170.000 0.000
joint 2 3 4 out of range

-90.000 -176.943 -44.277 71.220 170.000 -180.000
joint 2 out of range

-90.000 -176.943 -44.277 -108.780 -170.000 0.000
joint 2 out of range

Grading Criteria

- **Program (40%)**

- One-click execution (following the specified format)
- Accuracy, with two test cases provided (10% each)
- Clear comments (additional points for well-considered annotations)
- Check that rotation angles stay within the working range
- Neat output

- **Result Report (60%)**

- Interface description (development platform, execution instructions, etc.)
- Program structure explanation (program flow, core code explanations, etc.)
- Explanation and derivation of mathematical operations (please derive based on your code)
- Bonus: Discuss the pros and cons of two inverse kinematics methods (algebraic and geometric).

Notes

- Please present the result report as an electronic file in the format "Robotics_Project1_StudentID.pdf". The full score is 100 points; any score exceeding 100 will be capped at 100, with no extra points carried over to Project 2.
- Compress the complete project (code + result report) into a file named "Robotics_Project1_StudentID.zip."
- You may use any programming language to complete the project. If additional packages are required, please specify them in the report and provide clear instructions on how to execute the code.
- The mathematical operations in the result report can be presented using either handwritten scans or equation tools.
- The report must include:
- Name, student ID, and email (for contact if there are issues running the code)
- The following sections:
 - Program documentation (with screenshots of execution results)
 - Derivations of mathematical operations
 - References
 - Bonus topic (optional)

Additional Reminders

- Please do not plagiarize (TA will use moss to check).
- You cannot use existing robotic toolboxes for calculations.
- For mathematical operations, please use **atan2** to compute angles, as **atan2** has directionality and can avoid mathematical extremes.
- You can perform mutual verification of forward and inverse kinematics for this project, as the final project will require using this project to calculate kinematics. Therefore, it is essential to take this project seriously.
- If the target pose is within the working range, there will be eight sets of solutions without considering angle limits. All eight solutions must be displayed, and any joints that exceed the working range should be indicated as such.

Appendix

Robotics Toolbox for Python

[1] <https://github.com/petercorke/robotics-toolbox-python/tree/master>

Robotics Toolbox for Matlab

[2] <https://petercorke.com/toolboxes/robotics-toolbox/>